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Review

A review on machine learning-based precision agriculture techniques for crop farming monitoring with IOT

S. P. Sudha[1] · J. B. Shajilin Loret[1,2]

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Abstract

Precision agriculture, driven by advancements in machine learning (ML) and the Internet of Things (IoT), has revolutionized modern crop farming by enabling real-time monitoring, predictive analytics, and data-driven decision-making. Despite the growing integration of machine learning and IoT in precision agriculture, challenges such as data heterogeneity, sensor reliability, computational complexity, and cybersecurity continue to hinder optimal system performance. This paper addresses the need for robust, scalable, and secure ML-IoT frameworks to enhance real-time decision-making and sustainability in modern farming. This review explores the integration of ML techniques with IoT-based precision agriculture systems to enhance crop health monitoring, soil analysis, irrigation management, and yield prediction. Various ML algorithms have been extensively utilized for disease detection, nutrient optimization, and climate impact assessment. This paper examines the role of IoT sensors in collecting real-time data, such as temperature, soil moisture, and humidity, to facilitate precision farming. Furthermore, this study highlights challenges such as data heterogeneity, sensor reliability, computational complexity, and cybersecurity threats in ML-driven IoT frameworks. A comparative analysis of existing ML models, their accuracy, scalability, and computational efficiency is provided to evaluate their effectiveness in precision agriculture applications. Additionally, this review discusses the integration of edge computing and cloud-based architectures for optimizing data processing and decision support in smart farming. Future research directions focus on the development of hybrid ML models, explainable AI techniques, and blockchain-based secure data sharing for sustainable and scalable precision agriculture solutions.

Highlights

- Explore ML-IoT integration for crop health monitoring, soil analysis, irrigation management, and yield prediction.
- Examine IoT sensors' role in collecting real-time data to enhance precision farming efficiency.
- Discuss challenges, ML model comparisons, and future research on hybrid AI and blockchain for smart agriculture.

Keywords Precision agriculture · Machine learning · IoT · Crop monitoring · Smart farming · Deep learning · Yield prediction · Edge computing

✉ S. P. Sudha, sudhaaass@gmail.com; J. B. Shajilin Loret, shaji.jb20@gmail.com | [1]Department of Computer Science and Engineering, St.Joseph's College of Engineering, Chennai, Tamilnadu, India. [2]Department of Information Technology, Francis Xavier Engineering College, Tirunelveli, Tamilnadu, India.

Abbreviations

AI	Artificial intelligence
IoT	Internet of things
APPFMK	Adaptive privacy-preserving fuzzy multi-keyword search
DCNN-TL	Deep convolutional neural network with transfer learning
GIS	Geographic information systems
GPS	Global positioning system
LCA	Life cycle assessment
ML	Machine learning
NLCA	Novel lightweight cryptographic algorithm
PA	Precision agriculture
DL	Deep learning
PPE	Personal protective equipment
SOC	State of charge
UAV	Unmanned aerial vehicle
VRT	Variable rate technology

1 Introduction

Precision agriculture (PA), sometimes referred to as smart farming or precision farming, is a contemporary agricultural method that optimizes farming practices by utilizing cutting-edge technologies like sensors, drones, Geographic Information Systems (GIS), artificial intelligence (AI), and the Internet of Things (IoT). Precision agriculture allows farmers to accurately monitor, assess, and control field variability by using data-driven decision-making processes. PA customizes inputs like water, fertilizer, and pesticides based on real-time data from particular field zones, in contrast to conventional farming practices that administer uniform treatment across whole fields. Enhancing production, increasing resource efficiency, and reducing environmental effects are the main objectives of this strategy.

The focus of this study lies in exploring and reviewing the integration of machine learning (ML) techniques with Internet of Things (IoT)-based frameworks in the domain of precision agriculture. The primary motivation behind this work is driven by the necessity to address the global challenges associated with modern farming, including food security, climate variability, resource constraints, and the demand for sustainable agricultural practices. With agriculture being a resource-intensive sector, the application of ML algorithms for yield prediction, disease detection, irrigation management, and crop health monitoring offers promising solutions to optimize inputs and maximize outputs. Additionally, IoT systems provide real-time environmental sensing, which complements ML's predictive capabilities. This convergence forms the cornerstone of smart farming. The paper is motivated by the need to synthesize current developments, assess the effectiveness of various ML models, and highlight emerging technologies such as edge computing, blockchain, and AI-driven decision-making systems in transforming agriculture into a more efficient and resilient practice.

Technologies such as variable rate technology (VRT), GPS-guided machinery, and remote sensing help in precise application and monitoring, leading to more sustainable agricultural practices. The importance of precision agriculture in smart farming lies in its ability to address global agricultural challenges, such as climate change, food security, and resource scarcity. Farmers are under increasing pressure to produce more food with fewer resources as the world's population grows. Through intelligent irrigation systems, PA maximizes water use, minimizes overuse of pesticides and fertilizers, and makes sure crops get the proper quantity of nutrients at the correct time.

Through AI-powered picture identification and predictive analytics, it also aids in early disease diagnosis and pest control, lowering crop losses. AI and IoT technologies were utilized to enhance sustainable farming practices and optimize smart agriculture solutions [1]. Field monitoring is further improved by the combination of satellite imaging and unmanned aerial vehicles (UAVs), which provide high-resolution information on soil and crop health. By minimizing waste and maximizing efficiency, precision agriculture contributes to sustainable farming practices while ensuring higher yields and profitability. Table 1 and Fig. 1 displayed about the prediction of crop yield.

Machine learning techniques are categorized into supervised (e.g., Support Vector Machines, Random Forests), unsupervised (e.g., K-Means, DBSCAN), and reinforcement learning models based on their methodological paradigm. Additionally, a cross-sectional classification is provided based on data type—such as image data (used in computer vision for disease detection), time-series sensor data (applied in irrigation and climate modeling), and spatial data (derived from

Table 1 Crop yield prediction based on environmental factors

Crop type	Temperature (°C)	Soil moisture (%)	Fertilizer usage (kg/ha)	Predicted yield (tons/ha)	Actual yield (tons/ha)
Wheat	18.5	15.2	120	3.2	3.1
Rice	22.8	22.5	135	4.8	4.6
Corn	25.4	18.9	140	5.3	5.1
Soybean	20.2	14.3	110	2.7	2.6
Barley	17.0	13.8	100	3.0	2.9
Cotton	30.1	12.1	150	2.4	2.3

UAVs and satellites for soil and crop mapping). Furthermore, each model is contextualized according to its application domain within agriculture, including tasks like yield prediction, soil classification, disease diagnosis, pest monitoring, and resource optimization. This structured organization not only enhances the clarity and coherence of the review but also aids in comparative analysis and selection of appropriate models for specific agricultural challenges.

Given the rapid evolution of smart farming technologies and the emergence of new paradigms such as explainable AI, federated learning, and blockchain-based security protocols, there is a pressing need for an up-to-date review that reflects the latest advancements and practical implementations in precision agriculture. The agricultural landscape is experiencing a technological shift, and traditional surveys are no longer sufficient to capture the complexity and interdependence of contemporary tools. This review addresses the current gap in the literature by integrating state-of-the-art technologies, assessing recent deployments of ML models in real-world scenarios, and evaluating the incorporation of edge computing and secure data-sharing frameworks. Moreover, it consolidates insights on cost-effectiveness, scalability, and environmental sustainability, making it a timely contribution to guide researchers, policymakers, and practitioners toward the next generation of agricultural innovation. The necessity stems from the dynamic nature of agriculture's challenges and the transformative potential of emerging digital technologies.

Furthermore, precision agriculture plays a crucial role in automating farm operations and reducing labor dependency, which is particularly significant in regions facing agricultural workforce shortages. Autonomous tractors, robotic harvesters, and AI-driven decision support systems streamline operations, reducing human intervention while maintaining accuracy and productivity. Smart farming powered by PA also fosters climate-resilient agriculture by providing adaptive solutions to unpredictable weather patterns, helping farmers mitigate risks and enhance resilience.

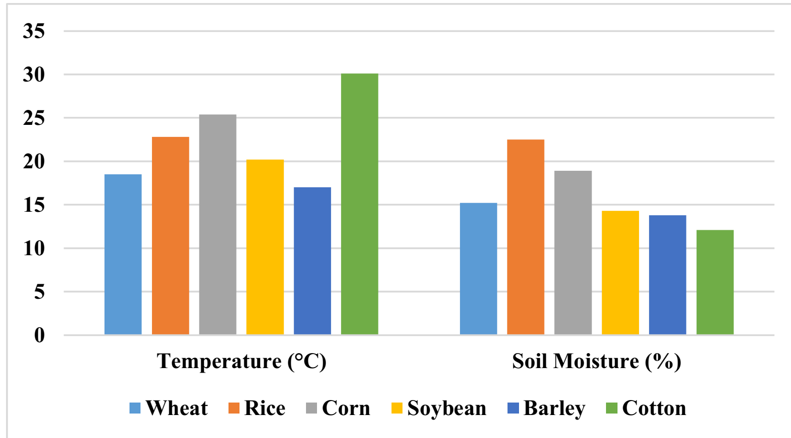
Machine learning techniques were utilized extensively to enhance decision-making in precision agriculture by offering intelligent data analysis tools. These tools improved various farming operations such as crop yield prediction, soil quality assessment, and pest detection. Algorithms like support vector machines, random forests, and deep learning models were applied to process multispectral images and sensor data. Integration with IoT and remote sensing technologies enabled real-time monitoring and responsive control systems. Challenges such as data heterogeneity and lack of standardization were highlighted, along with the need for scalable infrastructure. The review emphasized the transformative role of ML in achieving sustainable and profitable agriculture [2].

Supervised and unsupervised machine learning techniques were explored to support wireless sensor networks (WSNs) in precision agriculture. These techniques enabled efficient data fusion, anomaly detection, and predictive analytics for optimizing irrigation and nutrient management. Algorithms were embedded into low-power edge devices to support real-time decision-making and reduce network latency. The framework enhanced the adaptability of WSNs to dynamic environmental conditions, thus supporting site-specific agricultural interventions. A key contribution was the implementation of energy-aware models to sustain long-term operations. The study demonstrated how ML effectively managed and interpreted WSN data to improve agricultural productivity [3].

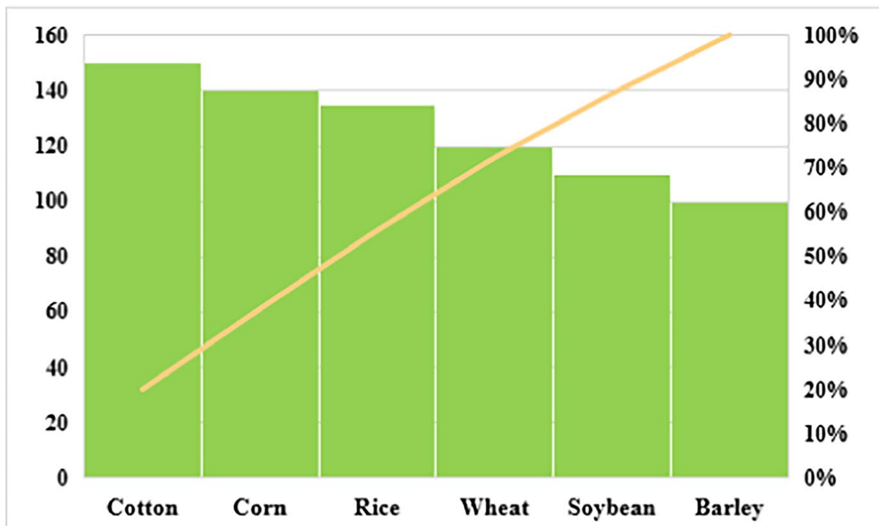
Precision agriculture practices were analyzed through the lens of machine learning integration, considering both technical and operational dimensions. The study discussed the utilization of data from satellites, UAVs, and in-field sensors to facilitate targeted applications such as variable rate application of fertilizers and pesticides. It outlined the implementation of classification and regression models for disease and weed detection. The practical considerations of infrastructure, cost, and farmer training were also emphasized. Emphasis was placed on enhancing crop quality and minimizing environmental impact through data-driven interventions. Overall, the work detailed a systematic roadmap from technological setup to field-level application [4].

Improved machine learning algorithms were developed to enhance precision farming through accurate modeling and robust data interpretation. The study focused on optimizing models for localized conditions, thus increasing their

Fig. 1 Crop yield production **a** temperature and moisture analysis **b** fertilizer usage **c** predicted and actual yield



(a)



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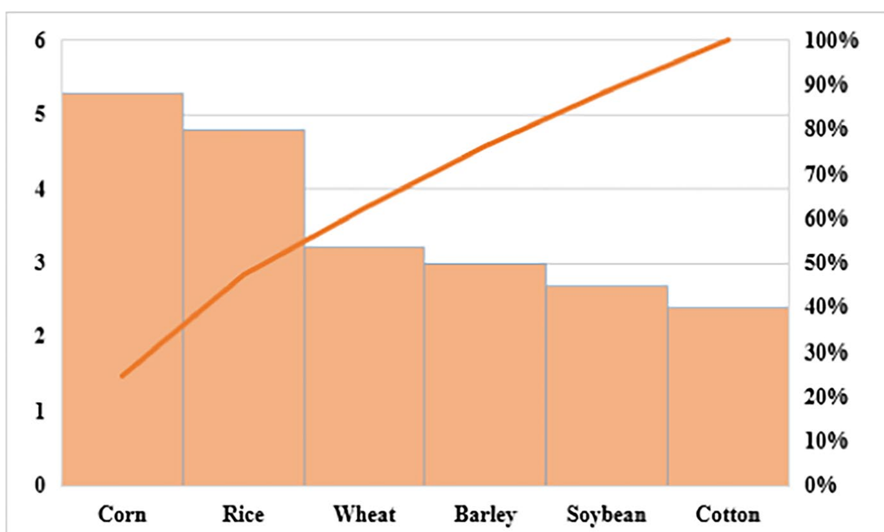

(c)

predictive accuracy. High-resolution imagery and sensor readings were utilized to detect plant health issues and soil variability. Feature engineering and model refinement contributed to the development of context-aware applications. These algorithms facilitated early warning systems and reduced dependency on manual inspections. The research highlighted the relevance of continuous learning models in dynamic agricultural environments [5].

A broad spectrum of machine learning applications in agriculture was reviewed, covering plant phenotyping, weed classification, and yield estimation. The study elaborated on the strengths and limitations of different ML algorithms like decision trees, neural networks, and ensemble models. Data sources included hyperspectral images, climate records, and in-field sensor data, which were used to train and validate models. The effectiveness of ML in supporting precision agriculture was tied to data quality and model interpretability. Moreover, the work emphasized collaborative frameworks involving agronomists and data scientists. It concluded with insights into future trends such as hybrid models and explainable AI [6].

Machine learning techniques were integrated with UAV technology to boost yield prediction and crop monitoring. UAVs provided high-resolution, multi-temporal data that was analyzed using supervised learning methods. The real-time feedback allowed for timely interventions in crop management, reducing input waste. The models demonstrated high accuracy in identifying stressed crops and assessing field variability. The study focused on adaptability to small and medium farms through cost-effective deployment. The overall system facilitated scalable yield optimization through intelligent data analytics [7].

An energy-neutral IoT system using machine learning was implemented for pest detection in agricultural fields. The design incorporated ultra-low-power hardware capable of continuous environmental monitoring and image capture. ML models were trained locally to identify pest activity with minimal energy consumption. The system used solar energy and optimized power usage for long-term sustainability. This innovation enabled autonomous operations in remote locations without infrastructure dependency. The approach contributed to efficient pest management with reduced operational costs [8].

Advanced machine learning methods were reviewed for their effectiveness in detecting biotic stress in crops, including diseases, pests, and fungal infections. Spectral imaging and phenotyping data were processed using classifiers like SVM, random forests, and neural networks. Feature extraction techniques improved the accuracy of early stress detection. The study highlighted multi-sensor integration for robust model development. Model performance was evaluated across diverse crop types and environmental conditions. The findings supported ML's potential in enhancing precision crop protection strategies [9].

The integration of IoT with machine learning in precision agriculture was explored, focusing on real-time monitoring and decision-making. Data from soil sensors, weather stations, and crop images were analyzed for actionable insights. ML models enabled predictive analytics for irrigation, fertilization, and disease management. Edge computing was utilized to reduce data transmission and processing delays. The study emphasized user-friendly interfaces to aid farmer adoption. The system's performance highlighted the synergy between IoT and ML in sustainable agriculture [10].

K-means clustering was employed within a dynamic IoT framework to enhance precision agriculture applications. The algorithm grouped sensor data based on similarities in soil moisture, temperature, and crop condition. This facilitated region-specific interventions and optimized resource allocation. The system adapted dynamically to changing field conditions by retraining clusters periodically. Communication among IoT nodes was improved through adaptive network configurations. The implementation demonstrated increased efficiency and decision accuracy in precision farming operations [11].

Machine learning and artificial intelligence were projected as the future foundation for precision agriculture in India. Emphasis was placed on affordability and scalability of intelligent systems for smallholder farmers. AI-driven models were used for yield prediction, disease identification, and climate adaptation strategies. Localized data was leveraged to train models that accounted for regional soil and weather variability. The study underscored the potential of digital agriculture in addressing food security. It called for public–private partnerships to bridge the technology gap in rural farming communities [12].

Spatial clustering techniques based on machine learning were applied to identify patterns in agricultural field data. Clustering helped differentiate soil types, crop growth stages, and nutrient variability zones. Algorithms were fine-tuned to handle high-dimensional spatial datasets generated from satellite and drone imagery. The study provided frameworks for automated field segmentation and precision treatment allocation. These methods improved operational efficiency and reduced input costs. The work showcased the role of spatial analytics in enabling data-driven agriculture [13].

Big data analytics was discussed as a key driver for decision-making in precision agriculture. The study focused on integrating large datasets from heterogeneous sources such as drones, satellite imagery, and IoT sensors. Machine learning algorithms were applied for trend analysis, anomaly detection, and performance forecasting. The real-time insights enabled timely interventions and optimized agricultural planning. The importance of data governance and infrastructure readiness was also addressed. The research highlighted the transition of agriculture towards a data-centric paradigm [14].

Precision agriculture practices were explored in detail, highlighting the journey from initial considerations to field applications. ML algorithms supported automated irrigation, fertilization, and crop health monitoring systems. Real-time feedback loops and historical data analysis facilitated adaptive decision-making. The study examined practical issues such as equipment calibration and connectivity limitations. Environmental and economic benefits were quantified through comparative analyses. The findings validated the operational effectiveness of machine learning in improving agricultural sustainability [4].

Emerging trends in computer vision and machine learning for precision agriculture were systematically reviewed. Techniques like object detection, semantic segmentation, and anomaly recognition were applied to monitor crop health. Convolutional neural networks and transfer learning were leveraged for disease detection and yield estimation. Integration with UAVs and mobile platforms allowed scalable field deployment. The study also addressed data annotation challenges and proposed hybrid learning solutions. The findings illustrated the growing impact of visual analytics in smart farming [15].

Unsupervised machine learning algorithms were surveyed for their effectiveness in processing agricultural datasets. Algorithms such as k-means, DBSCAN, and hierarchical clustering were tested on soil, crop, and weather data. These techniques enabled the discovery of latent patterns and trends without labeled data. Applications included crop classification, land use segmentation, and seasonal analysis. The study highlighted the benefits of reducing human intervention in model training. Results supported the use of unsupervised methods for exploratory analysis and decision support [16].

Algorithmic rationality in precision agriculture was examined to understand the influence of automated decision systems on farming practices. The study critiqued the notion that algorithms offer neutral, objective insights. It explored how data models shape perceptions of productivity, efficiency, and sustainability. The socio-technical implications of delegating decisions to algorithms were analyzed. Farmers' trust in data-driven tools and the risk of overreliance on automation were discussed. The research encouraged critical engagement with machine learning systems in agricultural settings [17].

Machine learning was applied to estimate agricultural waste and support sustainability initiatives. Models were developed to quantify surplus production, post-harvest losses, and inefficiencies in supply chains. Data sources included field sensors, harvest records, and market trends. The study proposed decision-support tools that advised on waste reduction strategies and resource optimization. Integration with circular economy principles was emphasized. The approach demonstrated the potential of ML in enhancing the sustainability profile of precision agriculture [18].

A robotic weed control system based on computer vision and machine learning was implemented for precision agriculture. Cameras captured real-time field imagery, which was processed using classification algorithms to detect weeds. The robotic arm targeted and removed weeds without affecting crops. The system was designed for energy efficiency and continuous operation in open fields. Edge processing enabled low-latency decision-making. This solution showcased the potential of automation in reducing manual labor and herbicide usage [19].

Applications of artificial intelligence in precision agriculture were broadly discussed, focusing on crop forecasting, yield optimization, and risk assessment. AI models synthesized data from weather sensors, market analytics, and crop records. Predictive modeling supported proactive decision-making to mitigate environmental and economic risks. The study emphasized farmer-centric design for technology adoption. Cloud platforms and mobile applications were used to deliver insights to end users. The research illustrated AI's transformative role in modern agricultural practices [20].

The novelty of this review lies in its holistic and multidisciplinary approach to precision agriculture, which not only surveys the conventional ML and IoT frameworks but also integrates the latest advancements in edge computing, blockchain, AI-based image processing, and predictive analytics. Unlike prior works, this paper uniquely focuses on the synergistic impact of combining IoT-enabled real-time data acquisition with intelligent decision-making powered by machine learning. It also includes an in-depth analysis of temporal data, environmental cost assessments, and data augmentation strategies that are rarely discussed in existing literature. Moreover, the inclusion of security protocols like homomorphic encryption, federated learning, and lightweight cryptographic models adds a novel dimension to the review, making it relevant for both research and practical implementation. By presenting a unified framework that spans technical, operational, and security aspects, the paper establishes a novel benchmark for future studies and smart farming applications.

The existing literature comprises numerous survey studies that have individually explored various facets of precision agriculture, including the application of IoT, machine learning algorithms, and automation techniques. However, many of these preceding reviews are either limited in scope—focusing solely on a specific technology—or outdated in terms of current innovations in cloud-edge integration and AI security frameworks. For example, earlier works reviewed ML models for yield prediction but did not include comparative analyses of recent deep learning models or consider the cybersecurity concerns posed by IoT systems. Other studies emphasized IoT deployment in agriculture but lacked integration with machine learning-driven analytics. This paper attempts to bridge these gaps by offering a comprehensive survey of both technological pillars—ML and IoT—and their collaborative impact on crop health monitoring, predictive analytics, and decision-making frameworks. The limitations identified in previous works, such as inadequate consideration of data privacy, lack of hybrid model discussions, and under-representation of real-time edge computing solutions, are critically analyzed and addressed in this study.

The structure of the paper is organized as follows: Sect. 2 presents a detailed review of the role of artificial intelligence in sustainable farming and smart agricultural systems. Section 3 focuses on specific ML algorithms applied in crop health monitoring and yield estimation. Section 4 highlights the key challenges and limitations in adopting ML and IoT frameworks. Finally, the paper concludes in Sect. 5 with a summary of findings and outlines future research directions.

2 Impact of artificial intelligence on sustainable farming

Precision agriculture is revolutionized by artificial intelligence (AI), which uses data-driven decision-making to increase production, sustainability, and efficiency. AI-powered systems evaluate real-time data gathered from several sources, such as drones, satellites, and ground-based sensors, by utilizing cutting-edge technologies like machine learning, computer vision, and the Internet of Things (IoT). IoT-based systems were implemented to improve precision farming by enabling automated monitoring and control of agricultural activities [21]. By monitoring soil quality, forecasting weather patterns, adjusting irrigation schedules, and identifying plant illnesses early on, these technologies help farmers save resources and boost crop yields. Precision planting and fertilization are made possible by AI-driven predictive analytics, which guarantees that crops get the appropriate nutrients in the right amounts at the right times, reducing environmental impact and increasing yield. AI-based image recognition techniques facilitate early pest detection and disease management by analyzing plant health through multispectral and hyperspectral imaging, leading to timely interventions. Precision agriculture techniques were integrated with IoT to enhance real-time data collection, leading to optimized resource utilization and increased crop yields [22]. AI-powered market analysis tools also help farmers make better decisions on supply chain management, demand forecasts, and crop pricing, which increases profitability. By combining artificial intelligence (AI) with intelligent irrigation systems, water distribution may be automated in response to current soil moisture levels, guaranteeing sustainability and effective water use. Additionally, by tracking animal health, identifying behavioral abnormalities, and anticipating disease outbreaks, AI improves livestock management and raises farm output overall. AI's contribution to precision agriculture is growing as it develops further, propelling developments in automation, resource optimization, and sustainable farming methods that eventually support environmental preservation and global food security.

2.1 IoT-driven smart agriculture: decision-making framework and real-time monitoring

This section illustrates an advanced IoT-based smart agriculture monitoring system that integrates multiple cutting-edge technologies, including wireless sensor networks (WSNs), drones, IoT-enabled tractors, and real-time data analytics, to optimize agricultural productivity and resource management. An IoT framework was proposed for precision agriculture, enabling real-time environmental monitoring and efficient farm management [23]. The system employs a network of sensors that continuously gather essential environmental and soil parameters. These sensors relay data to WSN nodes, which facilitate seamless communication between on-ground equipment and cloud-based data processing systems. Drones play a critical role in aerial monitoring, capturing high-resolution images and multispectral data for crop health assessment, pest detection, and precision spraying. IoT tractors further enhance automation by dynamically adjusting agricultural operations such as plowing, planting, and irrigation based on real-time sensor feedback. IoT-driven smart agriculture solutions were developed to enhance productivity, resource efficiency, and environmental sustainability in farming operations [24]. The integration of these components minimizes human intervention, improves operational efficiency, and ensures sustainable agricultural practices by optimizing the use of water, fertilizers, and pesticides.

Table 2 offers a comprehensive evaluation of IoT component affordability and economic viability for precision agriculture, particularly for smallholder farmers. Key parameters such as 5-year ownership cost and estimated return-on-investment (ROI) periods provide deeper insight into the long-term financial implications of adopting these technologies. Low-cost components like temperature and humidity sensors demonstrate high affordability and short ROI periods

Table 2 Comparative cost analysis of IoT deployment in smart farming

IoT component	Unit cost (USD)	Quantity per field	Total cost (USD)	Installation cost (USD)	Annual maintenance cost (USD)	5-Year ownership cost (USD)	Estimated ROI period (Years)	Affordability index*
Soil moisture sensor	45.0	20	900.0	150.0	100.0	1,550.0	2.5	Medium
Temperature sensor	35.0	15	525.0	100.0	80.0	1,025.0	2.0	High
Humidity sensor	30.0	15	450.0	100.0	75.0	975.0	2.1	High
Camera module	120.0	5	600.0	200.0	150.0	1,550.0	3.5	Medium
Drone (per unit)	1,500.0	2	3,000.0	500.0	400.0	5,500.0	4.5	Low
Gateway device	200.0	2	400.0	100.0	90.0	950.0	2.3	Medium

(~2 years), making them ideal for widespread deployment in resource-constrained farming environments. Soil moisture sensors and camera modules offer moderate affordability with balanced performance and cost–benefit returns. Conversely, high-cost technologies such as drones, despite their advanced capabilities, exhibit a long ROI period (4.5 years) and a high 5-year ownership cost (~USD 5,500), posing financial challenges for individual adoption. These findings underscore the importance of prioritizing economically scalable solutions and exploring shared or subsidized models for expensive assets, ensuring broader accessibility and sustainable technology integration in agriculture.

The potential of this smart agricultural system to use cloud computing and artificial intelligence (AI) for predictive analytics and real-time data processing is one of its primary features. The gathered information is sent to a central data center, where machine learning algorithms examine trends to predict possible hazards including crop diseases, insect infestations, and drought. An intelligent irrigation system was designed using IoT, blockchain and machine learning to optimize water usage based on real-time environmental data [25].

Real-time monitoring dashboards allow farmers to access critical insights via mobile or web-based applications, enabling them to make informed decisions remotely. Figure 2 and Table 2 analysed the cost analysis. The system also employs automated alert mechanisms that notify farmers of deviations from optimal conditions, allowing immediate corrective actions to be taken. By integrating IoT, AI, and WSN technologies, this intelligent decision-making framework enhances precision agriculture, boosts yield, and reduces resource wastage, making it a cornerstone for the future of sustainable and technology-driven farming.

2.2 Optimization process of precision farming

The IoT devices have revolutionized the way real. Agriculture's use of time data gathering and analysis integrating IoT with sensors. Due to technological advancements in farming operations, agricultural stakeholders may now obtain critical information on environmental parameters, health of the soil, and crop status quickly. Deep learning models were employed with IoT-enabled sensors for early detection and classification of crop diseases, improving agricultural productivity [26]. For instance, the use of soil moisture and climate sensors enables farmers to monitor the growth of crops actively and make decisions on time, therefore: optimizing resources and improving yields which was analysed int Table 3 and Fig. 3.

This integration not only facilitates immediate data analysis but also cultivates predictive capabilities, adding to the overall crop management. Cameras and drones are widely capturing high-resolution images of crops. A real-time soil moisture monitoring system was developed using wireless sensor networks and machine learning to optimize irrigation strategies [27]. Then, machine learning analyzes those images taken models which identify pests from visual features such as color, texture, and Motion.

With the use of computer vision, ML algorithms can detect specific identify pests on crops and classify them according to species/stage of infestation, which enables targeted control strategies. Recent 2works demonstrate that IoT combined

Fig. 2 Cost analysis

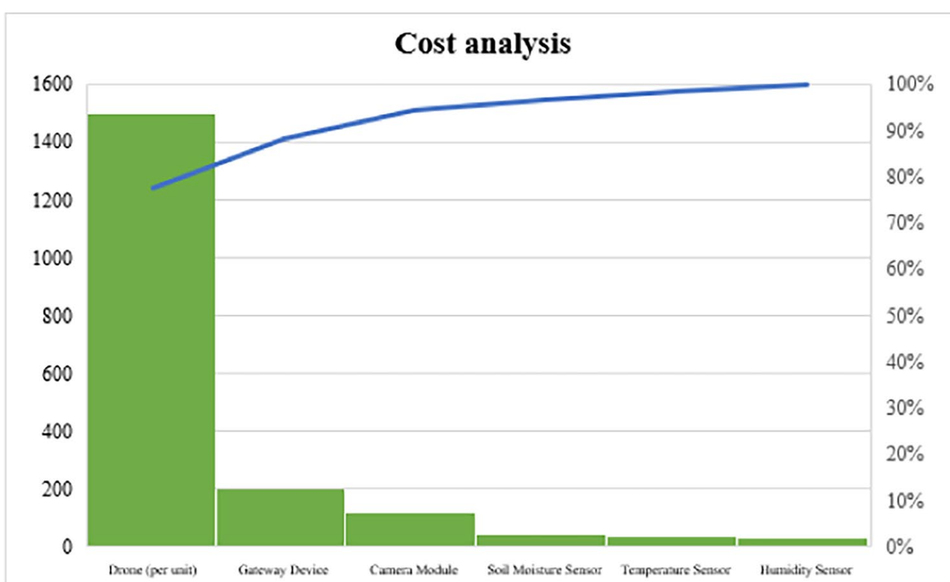

Table 3 Temporal analysis of key agricultural parameters for precision farming optimization

Time (Months)	Soil moisture	Temperature	Pest conditions	Crop productivity	Alert threshold
0	0.5	1.5	0.2	0.3	0.25
1	1.0	1.3	0.4	0.35	0.25
2	1.5	1.1	0.6	0.4	0.25
3	1.3	0.9	0.8	0.45	0.25
4	1.0	0.7	1.0	0.5	0.25
5	0.7	0.5	1.0	0.45	0.25
6	0.3	0.3	0.9	0.4	0.25
7	0.0	0.1	0.8	0.35	0.25
8	-0.2	-0.1	0.6	0.3	0.25
9	-0.3	-0.2	0.5	0.35	0.25
10	0.1	0.0	0.4	0.4	0.25

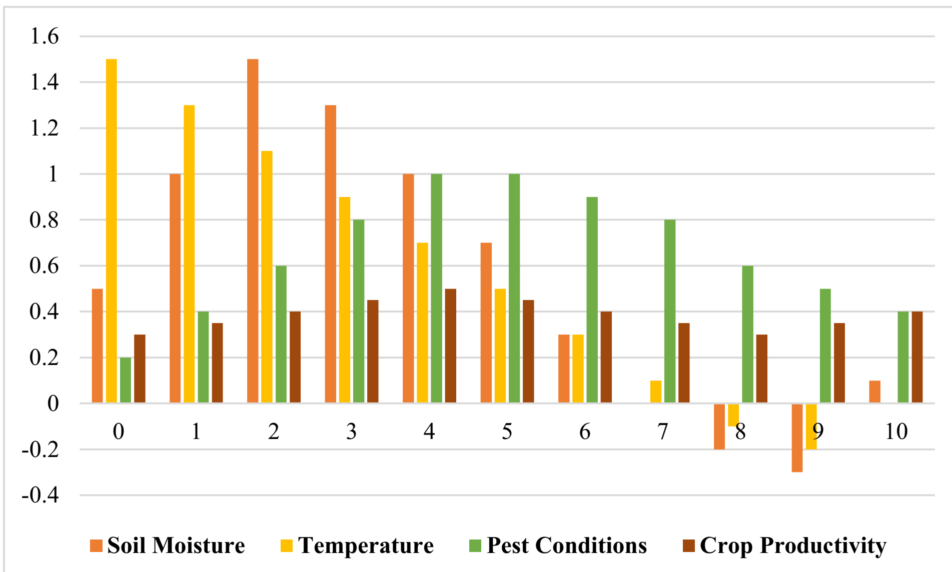

Fig. 3 Temporal analysis of crops

with cloud technology paves the way for sophisticated data analytics, which can provide information to farmers on trends and growth patterns, especially of sensitive crops like onions contribute to a more sustainable and productive farming environment. IoT and machine learning techniques were applied to develop precision fertilization models, ensuring optimal nutrient delivery for crop growth [28]. Further, artificial intelligence integrated with IoT systems serves to refine these processes, which reiterates that an interdisciplinary approach to sustainability is essential practices.

3 Part of machine learning in precision agriculture

By facilitating data-driven decision-making, streamlining resource allocation, and enhancing crop output forecasts, machine learning (ML) has completely transformed precision agriculture. Farmers can evaluate enormous volumes of real-time data from several sources. These data-driven insights help in monitoring soil health, detecting nutrient deficiencies, and predicting potential pest infestations with high accuracy [28]. Data analytics and machine learning were utilized to predict agricultural yield, improving decision-making and resource allocation in precision farming [29]. By leveraging supervised and unsupervised learning models, farmers can classify different soil types, assess moisture levels, and implement site-specific crop management strategies. Additionally, predictive analytics powered by ML algorithms can provide early warnings for diseases and unfavorable environmental conditions, allowing proactive intervention and minimizing crop losses.

Another critical application of machine learning in precision agriculture is yield prediction and crop planning. Traditional methods of estimating crop yield were largely based on historical trends and expert judgment, which often lacked accuracy and efficiency. However, with ML models like deep learning and ensemble techniques, vast datasets—including past yield records, soil composition, weather parameters, and genetic information—can be processed to generate precise yield forecasts. These predictions help farmers optimize their planting schedules, select the best-performing crop varieties, and manage input costs effectively. Moreover, reinforcement learning techniques are being increasingly adopted to refine agricultural decision-making by continuously learning from environmental variables and farming outcomes. This approach aids in automating decision support systems, reducing dependency on human expertise, and ensuring higher productivity with minimal resource wastage.

Smart irrigation systems powered by artificial intelligence can autonomously regulate water distribution, preventing both overwatering and underwatering, thus improving water use efficiency. Drone-based image analysis integrated deep learning techniques for automated weed detection, enhancing efficiency in agricultural field management [30]. Moreover, ML-driven remote sensing technology allows farmers to monitor large agricultural fields without manual intervention, significantly reducing operational costs. A cloud-based IoT platform was implemented to monitor and control precision agriculture operations, enabling real-time decision-making [31]. The integration of ML with Geographic Information Systems (GIS) further enhances mapping capabilities, enabling precise detection of drought-prone areas, soil erosion patterns, and other critical landscape characteristics that impact agricultural productivity.

Machine learning also contributes to precision agriculture by improving pest and disease management through real-time detection and classification systems. Advanced image recognition techniques using CNNs allow automated identification of pests, diseases, and weeds from high-resolution images captured by drones or smartphone cameras. Edge computing was integrated with IoT to enhance real-time data processing in smart farms, reducing latency and improving operational efficiency [32]. These AI-powered detection systems enable early diagnosis, reducing dependency on excessive pesticide usage and promoting sustainable farming practices. Furthermore, ML-driven robotic systems equipped with computer vision can be deployed in fields to perform precision weeding, targeted pesticide application, and autonomous harvesting.

3.1 Machine learning algorithms for crop health and yield prediction

Machine learning algorithms play a crucial role in crop health and yield prediction by analyzing vast amounts of agricultural data, including soil quality, weather conditions, pest infestations, and plant health parameters. Techniques such as Random Forest, Support Vector Machines, Artificial Neural Networks, and Convolutional Neural Networks process data collected from remote sensing, IoT-enabled sensors, and satellite imagery to assess growth patterns, detect diseases, and estimate yields with high accuracy. Regression models like Linear and Polynomial Regression help in predicting yield based on historical data, while deep learning models extract spatial and temporal features from multispectral and hyperspectral images to enhance precision. Machine learning techniques were applied for automated pest detection and classification, enabling early intervention in precision agriculture [33]. Clustering algorithms such as K-Means and DBSCAN group similar crop conditions, enabling targeted interventions and resource optimization. Recurrent Neural Networks and Long Short-Term Memory networks analyze time-series data for forecasting future agricultural outcomes, assisting farmers in making proactive decisions. Hybrid models combining multiple algorithms improve reliability by integrating different aspects of environmental and agronomic factors. Wireless sensor networks were deployed for environmental monitoring in greenhouse agriculture, ensuring optimal climate control for plant growth [34]. Reinforcement learning techniques further optimize decision-making in precision agriculture by continuously learning from real-time data inputs and adjusting predictions based on environmental dynamics.

The integration of optimization General algorithms, mainly chaos system-based ones, using computer vision and applied to the type of soil. The classification represents a milestone in smart farming technologies. This literature review now explores the use of the Chaotic Jaya Optimization Algorithm (CJOA) for Optimizing soil type classification systems based on computer vision. IoT and blockchain technologies were combined to enhance secure data management and traceability in precision agriculture applications [35]. This review investigate the development of optimization algorithms and the use of computer vision in soil classification. This in turn, will allow identifying how chaotic systems can play a privileged role in enhancing farming accuracy and efficiency operations. Soil-type classification has a great influence in precision agriculture by enabling the Farmers are further enlightened on soil characteristics, which consist of texture, fertility, the ability of the soil to retain moisture, and its pH. Accurate classification leads to optimized crop selection, resource management, and fertilization practices. This information helps in enhancing crop yield, reduction of waste and enhancing Sustainability by adopting the right agricultural practices based on soil characteristics. Computer vision technologies involving the processing and interpretation of visual information have seen rapid agriculture development. Concerning the soil classification, the Computer Vision systems can analyse images from drones, satellites, and ground-based cameras to detect visual features that correlate with their properties of soil, such as texture, color, and composition. Machine learning algorithms processed remote sensing data to predict soil nutrient levels, aiding precision agriculture decision-making [36]. These features serve as input to the classification algorithms determining soil type agricultural fields.

3.1.1 Integration of edge computing and cloud architectures in smart farming

The integration of edge computing and cloud architectures in smart farming enhances efficiency, scalability, and real-time decision-making by leveraging distributed computing resources. Edge computing enables localized data processing at the farm level using IoT devices, sensors, and AI-driven analytics, reducing latency and bandwidth consumption by filtering and analyzing critical information before transmitting it to the cloud. This ensures immediate responses to environmental conditions, optimizing irrigation, fertilization, and pest control while minimizing resource wastage. The cloud complements edge computing by providing extensive storage, high-performance computing, and machine learning capabilities, facilitating long-term data analysis, predictive modeling, and large-scale farm management. A smart irrigation system was developed using IoT and fuzzy logic control to optimize water distribution based on real-time environmental data [37].

Edge devices process localized data, alerting farmers to urgent issues such as disease outbreaks or equipment failures, while cloud-based analytics refine historical and regional data to offer optimized planting schedules, yield predictions, and market forecasting. The combination of edge and cloud infrastructures enhances automation in agricultural robotics, allowing autonomous tractors, drones, and robotic harvesters to function efficiently with minimal human intervention. Predictive analytics and Blockchain models were utilized to forecast crop yields using time series data, improving agricultural planning especially soyabean traceability [38]. Security and data integrity are strengthened through blockchain-enabled cloud storage and encrypted edge processing, mitigating cybersecurity risks and ensuring compliance with agricultural data regulations. By integrating AI-driven edge computing with cloud-based analytical models, smart farming operations gain higher accuracy, improved resource utilization, and enhanced productivity, supporting sustainable agricultural practices and increasing food security through data-driven precision farming.

3.2 Enhancing security and data privacy in IoT-enabled agriculture

The rapid deployment of IoT devices in smart farming introduces significant security and data privacy challenges. Weak encryption mechanisms, outdated firmware, and insecure communication channels further exacerbate these vulnerabilities, making agricultural IoT systems prime targets for cybercriminals. Blockchain-based security models can ensure data integrity by providing a decentralized, tamper-proof ledger, while lightweight cryptographic algorithms can offer enhanced protection without overburdening resource-constrained IoT devices. Differential privacy techniques and homomorphic encryption can be employed to anonymize and process data without exposing individual records, preserving confidentiality while enabling valuable agricultural insight.

Figure 4 illustrates the integrated framework of Smart Agriculture, highlighting the pivotal role of data analysis and machine learning integration in optimizing agricultural practices. At the core lies a centralized system that synthesizes data from multiple sources to enhance decision-making. This system collects environmental and biological data using various types of sensors: thermal sensors measure ambient temperature to ensure crops remain within optimal growth conditions; pH sensors assess soil condition to monitor acidity or alkalinity, which directly influences nutrient availability; and moisture sensors support irrigation management by evaluating soil water content to prevent over- or under-watering. Additionally, nutrient sensors guide fertilization by detecting the presence of essential minerals in the soil, thereby aiding precise input application. Image analysis techniques are employed to monitor plant growth, providing visual insights into crop development and potential stress symptoms, while machine learning detection systems identify and manage pest control, enabling early intervention and reducing pesticide use. The centralized data analysis unit integrates this multi-modal input, utilizing machine learning algorithms to derive actionable insights, predict outcomes, and automate responses, making agriculture more sustainable, efficient, and data-driven. Additionally, edge computing solutions reduce the risk of data interception by processing information closer to the source, minimizing the need for external data transmission. Autonomous drones were deployed for crop monitoring and spraying applications, enhancing

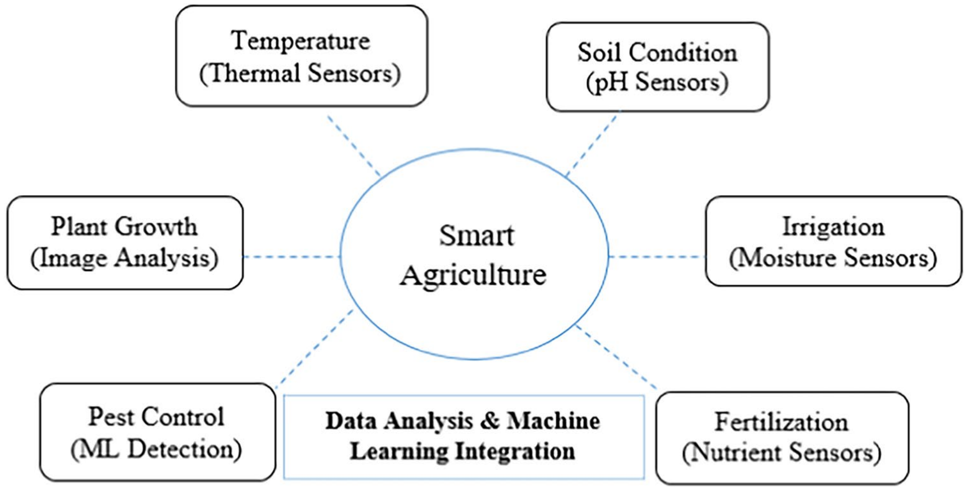

Fig. 4 IoT-based crop monitoring and data acquisition

efficiency in precision agriculture [39]. Secure data-sharing protocols, such as federated learning, allow multiple stakeholders to collaborate on predictive models without exposing raw data, safeguarding sensitive agricultural information from cyber threats.

The preprocessing and augmentation steps outlined in the table are tailored to accommodate the specific characteristics of different crops—rice, wheat, and maize—to enhance model performance and data representation which is shown in Table 4. Image resizing resolutions vary to capture critical features: rice images are resized to 800 × 600 pixels to emphasize finer leaf details, wheat to 1024 × 768 for overall plant structure, and maize to 1280 × 960 to encompass the larger cob features. The aspect ratio is uniformly maintained at 4:3 across all crops to preserve natural proportions. Color normalization employs mean RGB values and standard deviations optimized for rice and wheat ([123,117,104]) to maintain leaf color consistency, while maize uses slightly higher values ([125,120,105]) to accurately capture the yellowness of cobs. Random rotation angles differ by crop, with rice and wheat allowing ± 15° rotations to simulate natural wind-induced movement, whereas maize rotation is limited to ± 10° to prevent distortion of cob orientation. Horizontal flipping is applied at 50% probability for wheat and maize to increase dataset diversity but is avoided in rice due to leaf asymmetry. Brightness adjustments are set at ± 10% for rice to reflect subtle lighting changes, ± 20% for wheat, and up to ± 30% for maize, accounting for its greater exposure to variable outdoor lighting. Finally, Gaussian noise injection is applied uniformly with a noise level of $\sigma = 0.02$ across all crops to improve robustness against sensor noise and environmental variability. This crop-specific approach ensures preprocessing and augmentation techniques are optimally aligned with the unique visual features and growth conditions of each crop type.

3.3 Big data analytics transforming crop production through smart technologies

Big Data Analytics is revolutionizing crop production by integrating smart technologies that enhance precision, efficiency, and sustainability in modern agriculture. Machine learning algorithms process these vast datasets to detect anomalies, forecast disease outbreaks, and recommend precise interventions, reducing dependency on traditional farming practices that often lead to resource inefficiencies. Advanced predictive models assist in determining the best planting schedules, irrigation patterns, and fertilization strategies, ensuring optimal growth conditions while mitigating risks associated with climate variability.

The integration of cloud computing and edge analytics enables real-time monitoring of soil moisture levels, nutrient content, and pest infestations, empowering farmers with timely interventions that enhance productivity and reduce wastage. Automated machinery powered by AI-driven analytics enhances precision agriculture by adjusting seeding rates, spraying patterns, and harvesting techniques based on real-time field conditions, significantly improving operational efficiency. Blockchain technology further strengthens agricultural supply chains by ensuring transparency in food production and distribution, preventing fraudulent practices while enhancing traceability from farm to fork. The convergence of Big Data and smart technologies fosters sustainable farming practices by minimizing chemical overuse, optimizing water consumption, and reducing greenhouse gas emissions, thereby promoting environmentally responsible food production. An IoT-based smart greenhouse management system was designed with environmental control mechanisms to optimize plant growth conditions [40]. Deep learning models analyzed leaf images for automated plant

Table 4 ML data augmentation & preprocessing settings for crop image datasets

Preprocessing step	Parameter	Option 1	Option 2	Option 3	Option 4	Selected value	Augmentation technique	Crop-specific recommendation
Image resizing	Resolution (WxH)	640 × 480	800 × 600	1024 × 768	1280 × 960	1024 × 768	Bilinear interpolation	Rice: 800 × 600 for finer leaf details Wheat: 1024 × 768 for whole plant features Maize: 1280 × 960 for large cob structure
	Aspect ratio	4:3	16:9	1:1	3:2	4:3	-	4:3 for all crops to maintain natural proportions
Color normalization	Mean RGB values	[123,117,104]	[120,115,100]	[125,120,105]	[118,113,108]	[123,117,104]	Z-score scaling	Rice/Wheat: Use [123,117,104] for leaf color consistency Maize: Slightly higher means [125,120,105] to capture cob yellowness
	Standard deviation	[58,57,56]	[55,54,53]	[60,59,58]	[57,56,55]	[58,57,56]	Z-score scaling	Same as mean RGB
Random rotation	Angle range (°)	−10 to 10	−15 to 15	−20 to 20	−5 to 5	−15 to 15	Random rotation	Rice/Wheat: ± 15° to simulate wind movement Maize: ± 10° to avoid distorting cob orientation
Horizontal flipping	Probability (%)	0	25	50	75	50	Random horizontal flip	Rice: 0% (due to leaf asymmetry) Wheat/Maize: 50% to increase dataset diversity
Brightness adjustment	Intensity variation (%)	± 10	± 20	± 30	± 40	± 20	Random brightness adjustment	Rice: ± 10% for subtle changes Wheat: ± 20% Maize: ± 30% due to outdoor lighting variability
Noise injection	Noise level (σ)	0.01	0.02	0.03	0.04	0.02	Gaussian noise	Uniform across crops for robustness

disease classification, improving early disease detection and management [41]. This transformation is reshaping global agriculture, enabling small-scale and large-scale farmers alike to harness data-driven strategies that improve profitability, resilience, and food security while addressing the growing challenges posed by climate change and population growth.

3.4 Comparative analysis of machine learning models in crop monitoring

Comparative analysis in crop monitoring represents a dynamic interplay between cutting-edge technology and traditional farming practices. In this realm, diverse machine learning models emerge as critical tools that not only forecast yields but also diagnose plant health issues through real-time sensor data and remote imagery. Each model brings its own set of strengths and limitations to the table, requiring careful consideration based on application context, data availability, and operational constraints. Machine learning models applied in crop monitoring differ not only in predictive accuracy but also in feature interpretability, making model selection a critical task. Figure 5 and Table 5 analysed the comparative analysis of different ML techniques in precision agriculture. The comparative benchmarking of machin learning models for precision agriculture reveals significant variations in performance, computational efficiency, and contextual applicability. Convolutional Neural Networks (CNNs) achieved the highest accuracy (94.3%) and F1-score (93.1%) due to their ability to process high-resolution UAV imagery for pest and disease detection under variable open-field conditions, though at the cost of very high computational demand and training time (250 s). Long Short-Term Memory (LSTM) networks also performed well (92.7% accuracy) when trained on 15,000 time-series samples from rain-fed systems, effectively capturing temporal variability in irrigation control.

XGBoost (92.1%) and Gradient Boosting (91.8%) models demonstrated robust performance with high feature variability and were effective across semi-arid regions and multi-crop datasets, showing a good balance between accuracy and training efficiency. Random Forests offered high accuracy (90.2%) with moderate resource requirements and adaptability to mixed crop zones and diverse soil types. SVM achieved a respectable 87.5% accuracy with a high computational burden, trained on 10,000 samples from temperate climates and loamy soil conditions. Simpler models like k-NN and Naïve Bayes, with lower accuracy (85.4% and 80.3% respectively), proved suitable for resource-constrained or stable environments, such as controlled crop areas and greenhouses, due to their low computational costs and minimal training times. MLR, while the least accurate (78.9%), was effective in dryland scenarios with basic input features. This holistic evaluation underscores the importance of aligning model selection with data complexity, environmental variability, and operational constraints in real-world precision agriculture applications.

The success of machine learning in crop monitoring is heavily dependent on data quality, sensor precision, and preprocessing techniques [42]. Remote sensing data captured via drones and satellites require preprocessing steps such as noise reduction, band selection, and radiometric calibration before being fed into ML algorithms [43]. Ensemble learning techniques, including stacking and bagging, further improve model performance by aggregating multiple weak

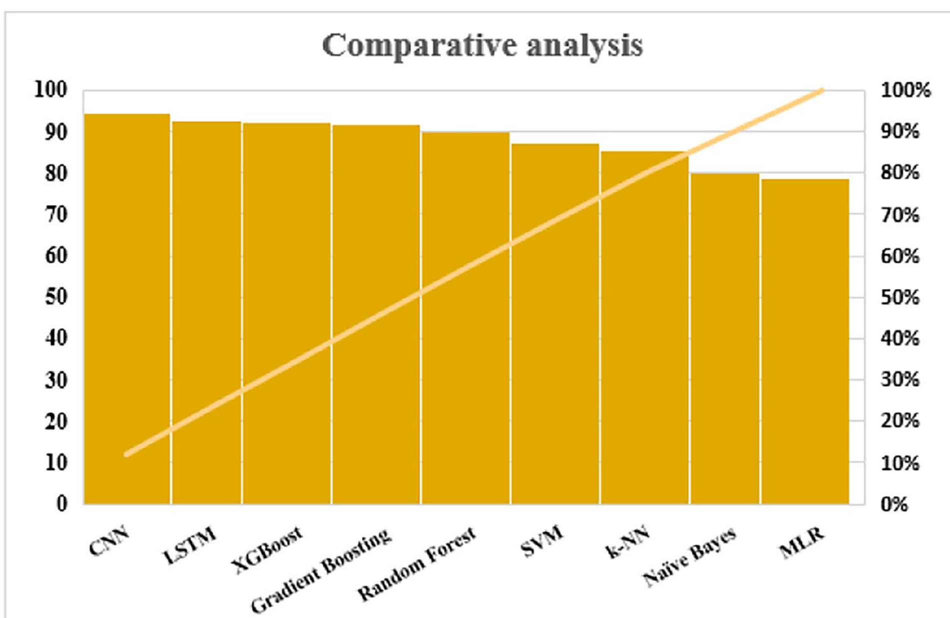

Fig. 5 Comparative analysis

Table 5 Comparative analysis of ML in precision agriculture

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Computational cost	Training time (s)	Inference time (ms)	Feature count	Dataset size	Feature variability	Environmental context
SVM	87.5	85.2	86.8	86.0	High	120	5	25	10,000 samples	Moderate	Temperate climate, loamy soil, 4 crop types
Random forest	90.2	89.7	89.1	89.4	Medium	80	3	30	12,500 samples	High	Mixed crop zones, seasonal rainfall, diverse soil types
Gradient boosting	91.8	91.1	90.6	90.8	Medium–High	95	4	35	11,000 samples	High	Varying pH levels, semi-arid conditions, multi-crop data
CNN	94.3	93.5	92.8	93.1	Very high	250	15	50	30,000 images	Very high	UAV-based pest/disease detection, open field variability
LSTM	92.7	92.0	91.4	91.7	High	200	12	45	15,000 time-series	High	Rain-fed systems, climate variability, irrigation control
k-NN	85.4	83.2	84.8	84.0	Low	30	2	20	6,500 samples	Low	Controlled crop area, minimal fluctuation
XGBoost	92.1	91.5	91.0	91.2	Medium–High	90	4	32	10,000 samples	High	Fertilizer-response tracking in multi-region trials
Naïve Bayes	80.3	79.1	78.5	78.8	Very low	15	1	15	5,000 samples	Low	Greenhouse-grown data, minimal environmental noise
MLR	78.9	77.5	76.8	77.1	Low	20	2	18	7,200 samples	Moderate	Dryland farming, basic weather and soil features

learners. Reinforcement learning was applied to optimize irrigation scheduling using IoT, ensuring efficient water usage and improved crop health [44]. However, trade-offs exist between model complexity and real-time inference speed. High-dimensional datasets often favor deep learning approaches, but in resource-constrained agricultural settings, lightweight models such as k-NN and Naïve Bayes may be preferred for on-field deployment.

Performance evaluation of machine learning models in crop monitoring extends beyond traditional metrics like accuracy and F1-score. Factors such as model interpretability, inference time, and hardware dependency significantly impact usability in real-world scenarios. While Random Forest and Decision Trees offer transparency in feature importance, deep learning models operate as "black boxes," requiring explainability techniques for decision justification. Moreover, the generalization ability of models varies based on climatic and geographical conditions. A CNN trained on a dataset from temperate regions may struggle when applied to tropical crops due to domain shifts. Transfer learning mitigates this challenge by adapting pre-trained models to new datasets. Evaluating the trade-offs between high-performing but computationally expensive models and lightweight but interpretable alternatives is essential for effective deployment in diverse agricultural landscapes.

4 Mathematical formulation

To strengthen the analytical framework of the proposed precision agriculture system, this section provides a set of comprehensive mathematical formulations that interconnect the domains of blockchain technology, cybersecurity, and edge–cloud computing. These formulations are not merely theoretical; they are applied directly to the context of smart farming and crop monitoring using IoT. By embedding these models into the domain of precision agriculture, the section enhances the practical relevance and serves as a guide for future implementations and research in agricultural technology systems.

In smart agricultural environments, maintaining the integrity and immutability of data generated by IoT devices—such as sensors measuring soil moisture, temperature, humidity, and crop health—is of utmost importance. Blockchain technology ensures data trustworthiness through a consensus mechanism, often implemented via the Proof of Work (PoW) algorithm. The PoW condition can be mathematically represented as

$$H\left(n\middle\|h_{prev}\middle\|T\right) < \text{Target} \tag{1}$$

Here, H is a cryptographic hash function (e.g., SHA-256), n is the nonce, h_{prev} is the previous block hash, and Trepresents the transaction data. Within the context of precision farming, this equation ensures that the collected data—such as irrigation history or pH levels—cannot be tampered with after being recorded, thereby enabling a secure and traceable data chain throughout the agricultural lifecycle.

Another essential component in IoT-driven agriculture is ensuring trustworthiness among thousands of interconnected edge devices deployed across the farm. Malfunctioning or compromised devices can lead to erroneous interpretations and poor crop management decisions. Therefore, a trust evaluation model is applied to continuously assess device reliability. The trust value *T_i(t)* of device *i* at time *t* is calculated using the equation

$$T_i(t) = \alpha \cdot T_i(t-1) + (1-\alpha) \cdot F_i(t) \tag{2}$$

where $T_i(t)$ is the trust value of device ii at time t, $\alpha \in [0,1]$ is a forgetting factor, and $F_i(t)$ is a feedback function based on behavior (e.g., successful authentication, anomaly detection). In a smart farming scenario, this model allows automatic identification of sensors that may be generating faulty readings, such as abnormal temperature spikes or inconsistent soil moisture levels, and isolates them to maintain system accuracy.

Cybersecurity plays a significant role in agricultural IoT networks, especially in defending against unauthorized access or cyber attacks that can disrupt farming operations. One widely used method in this domain is anomaly-based intrusion detection. It involves computing an anomaly score *A(x)* based on statistical deviations, given by the formula

$$A(x) = \frac{(x-\mu)^2}{2\sigma^2} \tag{3}$$

Here, x is the observed behavior (e.g., packet rate), μ is the mean, and σ is the standard deviation of historical data. A higher anomaly score A(x) indicates deviation from normal behavior and triggers alerts. In agricultural systems, this anomaly detection model can flag sudden changes in data transmission frequency from a sensor, which may indicate a denial-of-service attack or sensor malfunction, thereby enabling proactive threat mitigation to protect data integrity and continuity in decision-making.

To support efficient processing of agricultural data and reduce communication latency, a hybrid edge–cloud computing framework is often adopted. Determining where to process a given computational task—locally at the edge or centrally in the cloud—can be formulated as an optimization problem. The objective is to minimize latency while satisfying resource constraints and is mathematically expressed as

$$\begin{gathered}\min_{x_i \in \{0,1\}} \sum_{i=1}^{N} x_i \cdot D_i^{edge} + \left(1 - x_i\right) \cdot D_i^{cloud} \\ \sum_{i=1}^{N} x_i \cdot R_i \leq R_{max}^{edge}, \quad \sum_{i=1}^{N} \left(1 - x_i\right) \cdot R_i \leq R_{max}^{cloud}\end{gathered} \tag{4}$$

Here, x_i is a binary variable denoting if task ii is processed at the edge (1) or cloud (0), D_i^{edge} and D_i^{cloud} represent delay, and R_i is resource requirement. In precision agriculture, time-sensitive operations—such as pest detection via drone footage or automatic irrigation decisions—are best handled at the edge for real-time responsiveness, while more extensive tasks like trend forecasting or crop yield modeling can be relegated to the cloud.

Another important consideration in integrating blockchain into agricultural IoT is ensuring that the network can handle the high volume of sensor-generated data. The blockchain throughput *Θ* is an essential metric, indicating the number of transactions processed per unit time. It is estimated using the formula

$$\Theta = \frac{B \cdot T_S}{t_b} \tag{5}$$

where B is the number of transactions per block, T_s is the size of each transaction, and t_b is the average block generation time. For large-scale farms where multiple sensor readings are collected per second, this formula helps determine whether the blockchain infrastructure can scale efficiently without losing data fidelity or causing delays.

Finally, confidentiality of agricultural data—such as GPS coordinates, pesticide usage, or proprietary cultivation techniques—is critical. To evaluate the unpredictability and security strength of encrypted data, Shannon entropy is employed. The entropy *H(X)* of a dataset *X* is given by the formula

$$H(X) = -\sum_{i=1}^{n} p\left(x_i\right) \log_2 p\left(x_i\right) \tag{6}$$

where H(X) is the entropy of the data source X, and $p(x_i)$ is the probability of symbol x_i. Higher entropy values imply greater unpredictability and, hence, stronger data protection. This is especially important when transmitting sensitive data across public or semi-public networks between field devices and cloud storage platforms.

The mathematical models presented in this section provide a solid analytical basis for the integration of blockchain, cybersecurity, and edge–cloud computing into precision agriculture. By contextualizing these models within crop monitoring and IoT-based smart farming applications, this section not only adds depth to the proposed framework but also enhances its practical relevance for researchers and practitioners seeking to optimize agricultural productivity, security, and system scalability.

5 Challenges in machine learning and IOT-driven precision agriculture

The core issues and challenges hindering the widespread adoption of Machine Learning (ML) and IoT-driven precision agriculture are multifaceted.

1. Firstly, data management presents a significant hurdle, grappling with the sheer volume of sensor data that requires robust infrastructure for collection, storage, and real-time processing, particularly in often remote agricultural settings with limited or inconsistent connectivity.

2. Data quality and reliability are paramount; inaccurate or incomplete data can lead to skewed ML model outputs, resulting in flawed agricultural decisions and resource mismanagement.
3. Secondly, the development and deployment of contextually relevant ML models is complex, as agricultural environments are inherently diverse and dynamic, necessitating algorithms that can adapt to variations in soil composition, climate patterns, and crop-specific needs. The high cost of implementation acts as a substantial barrier, particularly for resource-constrained smallholder farmers, encompassing the acquisition of IoT sensors, data infrastructure, and specialized software.
4. Compounding this is the lack of technical expertise in many rural areas, necessitating extensive training programs and ongoing support to ensure effective utilization of these technologies.
5. Cybersecurity vulnerabilities pose a significant risk, as the interconnected nature of IoT devices makes them susceptible to malicious attacks, potentially compromising sensitive agricultural data and disrupting operations. Finally, interoperability challenges persist, hindering seamless data exchange and system integration between disparate IoT devices and platforms, creating silos of information and impeding the development of holistic precision agriculture solutions.
6. One critical challenge not addressed in the original analysis is class imbalance in crop disease datasets, which significantly affects the reliability of machine learning models in agricultural disease detection. In real-world datasets, healthy crop images often vastly outnumber diseased samples, leading to biased model training where classifiers disproportionately favor the dominant class. This imbalance reduces the model's sensitivity to rare but agriculturally important disease instances, resulting in poor recall and under-detection of early-stage infections.
7. To ensure robust performance, especially in disease surveillance and early warning systems, it is essential to apply techniques such as Synthetic Minority Over-sampling Technique (SMOTE), focal loss functions, or data augmentation strategies that artificially balance class distribution. Additionally, proper evaluation using metrics like class-wise precision, recall, and area under the ROC curve should be emphasized over overall accuracy to ensure fairness and reliability in predictions. Addressing class imbalance is thus fundamental for deploying trustworthy and actionable ML models in precision agriculture.
8. Another significant limitation of the paper is its superficial treatment of security and blockchain integration in IoT-based precision agriculture. While blockchain is briefly cited as a remedy for IoT vulnerabilities, the discussion lacks technical depth, omitting crucial aspects such as consensus mechanisms (e.g., Proof of Stake vs. Practical Byzantine Fault Tolerance), smart contract implementation, and the impact of blockchain latency on real-time decision-making. These omissions undermine the feasibility and scalability of proposed security enhancements in agricultural IoT systems.
9. Furthermore, the exploration of edge computing remains largely conceptual, without addressing key operational challenges critical to rural deployment—namely, limited memory capacity, the need for model compression techniques (such as pruning or quantization), and energy constraints in battery-powered edge devices. In low-resource settings typical of many agricultural environments, overlooking these constraints compromises the practicality of edge-AI solutions. For meaningful deployment, the review should critically evaluate trade-offs between local inference and cloud offloading, propose lightweight model architectures, and consider energy-aware scheduling techniques to ensure sustainable operation in field conditions.

6 Research gap

The existing studies in precision agriculture using machine learning and IoT technologies, while extensive and promising, reveal several critical gaps that necessitate further investigation. Firstly, many studies are limited by their focus on specific crops, regions, or environmental conditions, reducing the generalizability and scalability of their models across diverse agricultural landscapes. Secondly, although machine learning algorithms like SVM, CNN, and random forests have shown effectiveness, there is a lack of comparative studies that benchmark these models under uniform datasets to identify the most suitable algorithms for specific agricultural applications. Additionally, most research prioritizes technological development without adequately addressing the practical challenges faced by farmers, such as high implementation costs, lack of digital literacy, and limited access to infrastructure in rural areas. The integration of AI with real-time decision-making frameworks remains underexplored, especially in the context of edge computing and resource-constrained environments. Moreover, the issue of data quality—due to sensor malfunctions, noise, and inconsistencies in field data—poses a significant barrier to building robust predictive models. Studies often overlook the importance of explainability and interpretability in AI systems, which are essential for gaining user trust and ensuring transparency in decision-making processes. Lastly, ethical concerns, data privacy, and socio-economic implications of deploying autonomous systems in agriculture are seldom discussed, highlighting the need for interdisciplinary approaches that combine technological innovation with human-centered design and policy frameworks. Addressing these gaps will establish a comprehensive and inclusive foundation for advancing smart agriculture practices globally.

7 Conclusion

This review paper highlighted the transformative role of machine learning-based precision agriculture techniques in optimizing crop farming through IoT-enabled monitoring systems. By integrating real-time data acquisition, predictive analytics, and automated decision-making, these technologies significantly enhanced crop yield, resource efficiency, and sustainability. This study underscored the effectiveness of various machine learning models for soil health assessment, yield prediction, and disease detection. Furthermore, the fusion of IoT sensors with AI-driven analytics ensured continuous monitoring, early anomaly detection, and adaptive farm management strategies. Despite the evident advantages, challenges such as data security, high implementation costs, and model generalization across diverse agricultural landscapes persisted. Addressing these issues required collaborative efforts in developing cost-effective IoT solutions, robust machine learning frameworks, and scalable deployment strategies. All things considered, this study showed that precision agriculture powered by machine learning has enormous potential to transform contemporary farming and open the door to more sophisticated, data-driven, and sustainable farming methods.

7.1 Future scope

Future work should focus on the development and deployment of lightweight, interpretable machine learning models tailored for low-resource rural environments, addressing real-world constraints such as limited connectivity, sensor reliability, and heterogeneous data quality. The integration of reinforcement learning can be explored for adaptive irrigation and fertilization scheduling, where dynamic decision-making under uncertain weather and crop conditions is required; however, its effectiveness depends on availability of continuous feedback and robust reward structures. Similarly, generative AI, particularly generative adversarial networks (GANs), holds potential in augmenting limited agricultural datasets—especially for rare crop diseases—but requires careful domain adaptation to ensure realism and biological accuracy. Advancements in domain-specific transfer learning and federated learning could also enable scalable AI solutions while preserving data privacy. Future research must align these innovations with localized farming needs, economic viability, and user accessibility to ensure responsible, inclusive, and sustainable adoption of AI in precision agriculture.

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Declarations

Ethics approval and consent to participate This study does not involve human or animal subjects and, therefore, does not require ethical approval.

Consent for publication All authors provide their consent for the publication of this work.

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